Econ 5015-01
Computing and Machine Learning for Economics

Fall 2026 Syllabus

Instructor: Aric Shafran
Class Time: TR 12:00 PM -1:20 PM
Location: 03-207
Email: ashafran@calpoly.edu

Office Hours:
In-Person office hours: TR 1:30-2:45 PM, Room 03-340
Zoom office hours: You can schedule a 15-minute appointment for a Zoom meeting during my regularly scheduled office hours (typically Wednesday 1:00-2:00): https://scheduler.zoom.us/ashafran/officehours
By appointment: I am also available by appointment at other times. Please e-mail me to schedule an appointment.

Textbook:
The following books are recommended:

1. Introduction to Computation and Programming Using Python by Guttag, 3rd Edition, https://mitpress.mit.edu/9780262542364/introduction-to-computation-and-programming-using-python/
2. Python for Data Analysis: https://www.oreilly.com/library/view/python-for-data/9781491957653/
3. Python Data Science Handbook: https://jakevdp.github.io/PythonDataScienceHandbook/
4. An Introduction to Statistical Learning: https://www.statlearning.com/

Course Description:
This course provides an introduction to computer programming using Python. Although the course will be taught using Python, an emphasis will be placed on general rules and best practices for programming that are applicable to many programming languages. The first part of the course will cover general programming concepts including data structures, flow control, functions, object-oriented programming, and debugging. The second part of the course will apply this general programming knowledge to solve problems in data science and econometrics including data wrangling, web scraping, database

querying, model selection and cross validation, and implementing selected machine learning methods.

By the end of this course you should be able to:
1. Write computer programs in Python that use basic computer programming features including data structures, looping, and functions.
2. Use computers to transform large unorganized datasets into a useful format.
3. Write SQL queries to extract data from a relational database and import into Python
4. Write computer programs that implement a variety of economic and statistical methods including linear regression, cross validation, decision trees, and clustering.

How To Get Python (Anaconda):
Go to https://www.anaconda.com/download.
We will use Spyder and Jupyter, both of which are installed automatically with anaconda. You do NOT need to install PyCharm.
If you prefer to use a different distibution of Python or a different IDE, that is fine, but I will not offer any support for it.

Grading:
40% Weekly Assignments and other HW and in-class assignments
5% Quiz
20% Midterm
35% Final

Important Dates:
Quiz: TBD (Week 4 or Week 5)
Midterm: Thursday October 15 (in class)
Final: Thursday December 17 (1:00 - 3:30 PM)

Assignments:
The only way to learn to be a programmer is through practice. This is especially true because most of the time spent programming is debugging, and the more problems you have encountered in the past, the quicker you can catch future errors. With that in mind, this course will have weekly assignments. In addition to the weekly assignments, there will periodically

be short homework assignments and in-class assignments, some of which will be turned in. For many of the assignments, you can work in groups of 2, if you choose. I recommend working in groups of 2 when allowed, as it is often easier to debug a program with two people.

While it is OK to refer to static websites to get assistance, it is best to avoid these if you are copying and pasting without understanding the code. You should be able to complete all assignments using only course notes and textbooks.

Do NOT use AI-based websites that write the code for you. Turning in AI-generated code will result in a zero for that assignment and may be treated as scholastic dishonesty under university policy.

The weekly assignments will typically be due on Friday. Assignments submitted after the Friday deadline but within one week of due date will receive a 10% penalty. Assignments will not be accepted after one week past the due date.

Most assignments will be turned in via gradescope. Please register with my course by going to gradescope.com, then click "Sign Up", then click "Student", and then provide your info and the following course code: 6XR2W4. With gradescope, when you submit a python script, the site will automatically test your script, assign a grade, and notify you of how you did. You can resubmit as many times as you want prior to the deadline. Final grades for an assignment will be determined by the instructor based on results of the test-file as well inspection of code; final grades will not necessarily be the same grade assigned by gradescope.

General Policies:

- No make-ups will be given unless prior permission is obtained and only in extremely extenuating circumstances. If you miss an exam, you will be given a zero.
- I will make reasonable accommodations for persons with documented disabilities. Please notify me no later than the end of the second week of the term. For more information, visit http://drc.calpoly.edu.

- Cal Poly will not tolerate academic cheating or plagiarism in any form. (Visit https://academicprograms.calpoly.edu/content/academicpolicies/Cheating.)

Tentative Course Outline:
As the heading suggests, this outline is tentative. Prior to each exam, I will clearly define the material you will be responsible for.

Week 1:
Introduction to Python

Week 2:
Conditional Execution
Iterables and Looping

Week 3:
Algorithms
Functions
Mutability

Week 4:
Debugging
NumPy Arrays

Week 5:
Statsmodels
Pandas DataFrame

Week 6:
Objected-Oriented Programming
Dictionaries

Week 7:
Analysis of Algorithms

Week 8:
Data Wrangling
Midterm

Week 9:
Data Collection: API, XML

Week 10:
Data Collection: JSON, web scraping

Week 11:
Introduction to Relational Databases
SQL Queries

Week 12:
SQL Queries

Week 13:
Introduction to Statistical Learning Model Selection

Week 14:
Cross Validation
Decision Trees

Week 15:
Clustering
Catch-up and Review

Final Exam Week:
12/17: Final Exam 1:00-3:30 PM